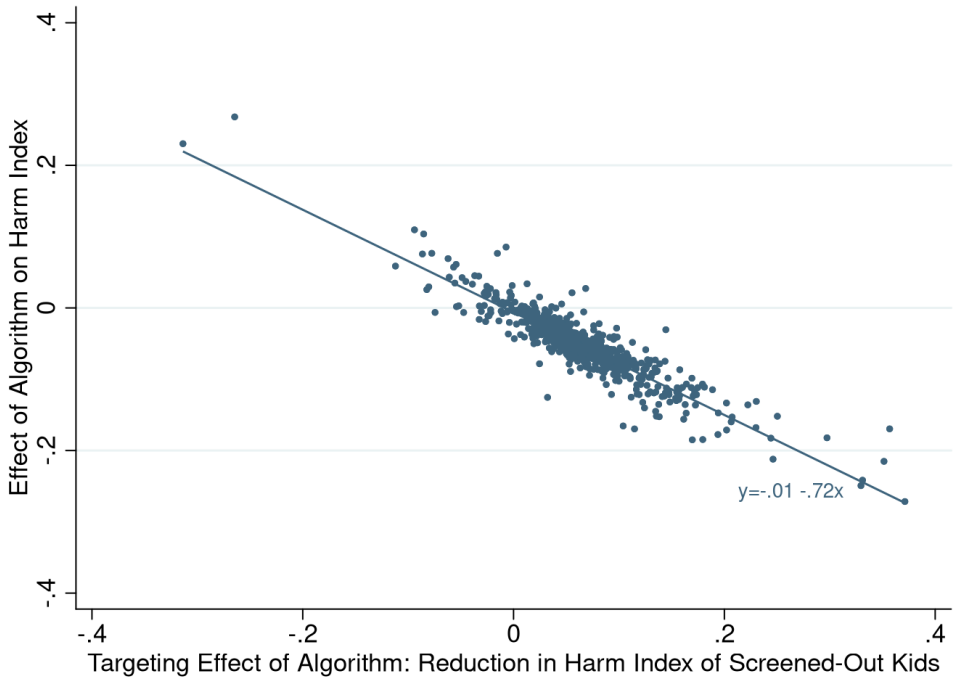
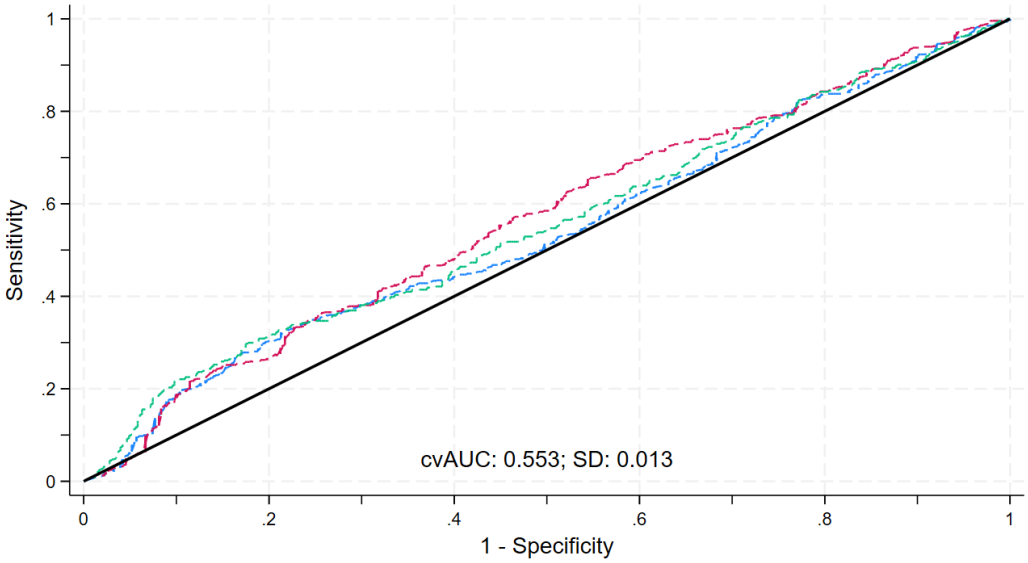
Figure A.14: Improvements in Targeting Associated with Reductions in Child Harm



Notes: We generate a thousand random groups of varying size in our data (minimum size 100 children). Within each group we then estimate effects of tool availability on targeting of screen-outs using the harm index (horizontal axis) and the effect of providing the tool on the harm index in the group (vertical axis). Each dot then represents effects of the tool on harm and targeting for each respective group. The horizontal axis measures targeting by leveraging the difference in harm after the referral among screened-out children with the tool compared to screened-out children without the tool (relying on random assignment to treatment and control, implying that the mean harm in treatment and control would have been similar in absence of treatment). This corresponds to the targeting test reported in Column 1 of Table 2. We plot a fitted line and report the coefficient estimates in the bottom right of the figure next to the fitted line.

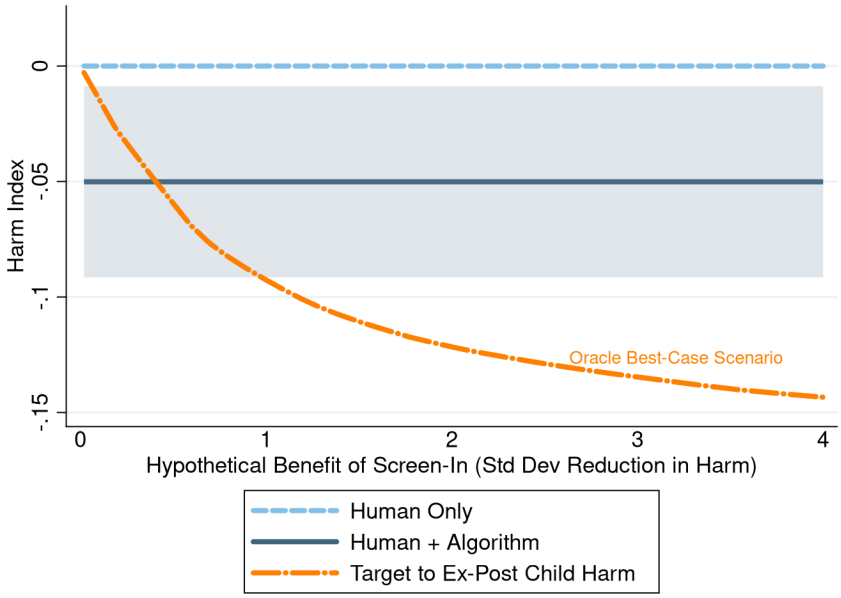
Figure A.15: Prediction of Algorithm Support from Discussion Text Data

Using 100+ Topics Generated from Word Embeddings by BERTopic



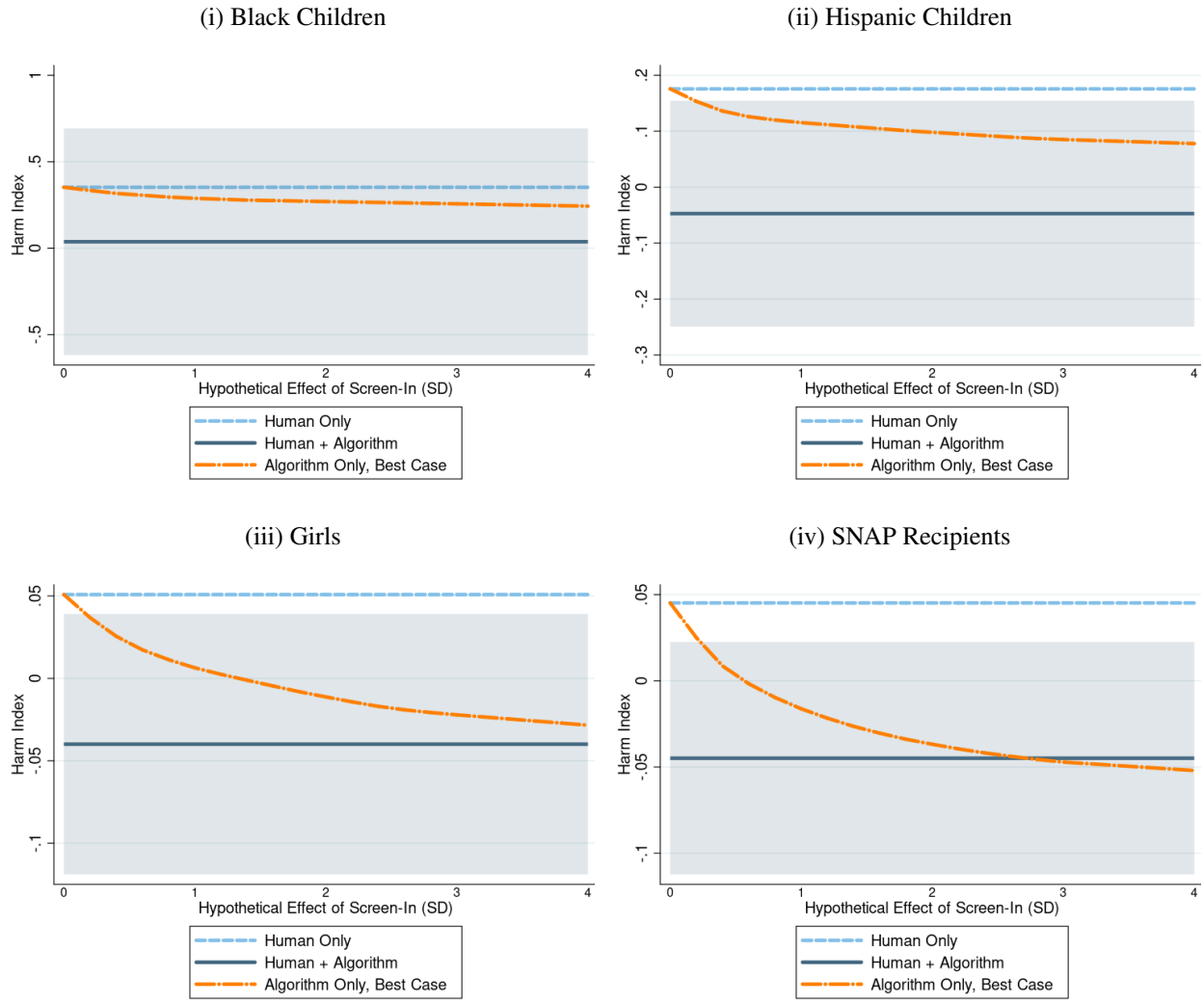
Notes: This figure shows the three-fold cross-validation ROC curves on the trial sample at the referral level since discussion notes are the same for all children on the same referral ($N = 2,668$). A small number of referrals from our sample that have no discussion notes are omitted from the analysis. The topics selected by the BERTopic model are used to predict whether workers had access to the algorithm tool for two sets of folds of the data, and are then validated on the remaining fold. The cross-validated mean AUC and cross-validated standard deviation are reported at the bottom of the figure. The BERTopic model was run on a set of eleven thousand referrals. It converts the text data using sentence transformers into word embeddings, which it then reduces dimensionality of using a state-of-the-art UMAP to avoid the curse of dimensionality when applying the HDBSCAN clustering algorithm for these embeddings. In practice, we used a seeded topic model to guide the model through some of the technical, profession-specific terminology used in the notes and ensure greater replicability across runs.

Figure A.16: Oracle Best-Case Decision Rule: Targeting Screen-Ins by Ex-Post Harm



Notes: This figure shows average child harm under three scenarios. The first scenario is a benchmark control group where humans do not have access to the algorithm (light blue short-dashed line). The second scenario is when humans have access to the algorithm, which is the experiment's intervention group, shown with a 90% confidence interval (solid dark blue line with shaded confidence bands; main estimates presented in Table 1). Finally, the figure shows results under a perfect ("oracle") decision-maker subject to the same screen-in rate, where screen-ins are targeted using ex-post harm under a range of assumptions about the benefits of screen-in (orange dashed-dotted line). The best-case scenario for the oracle assumes that the children who were screened in by humans, but for whom the oracle would have screened out, would have been just as well off under the oracle decision. Effects of this new decision rule depend on hypothetical effects of a screen-in on the child harm index (range of hypothetical screen-in benefits, in terms of standard deviation reductions in harm, shown on horizontal axis).

Figure A.17: Health Outcomes for Subgroups of Children with Algorithm-Only Counterfactual



Notes: For each child subgroup as reported in the title of each figure, we plot the average child harm for three different scenarios: when humans do not have access to the algorithm (light blue short-dashed line) which is our reference/control group, when humans have access to the algorithm (solid dark blue line, with shaded area representing a 90% confidence interval; main estimates presented in Table 1), and the best-case scenario for the algorithm (orange dash-dot line). The best-case scenario for the algorithm assumes that the children who were screened in by humans, but for whom the algorithm would have screened out, would have been just as well off under the algorithm decision. Effects of the algorithm screening in children whom humans screened out depends on the hypothetical benefits of a screen-in on the child harm index (horizontal axis). SNAP (food stamp) eligibility is a proxy for lower socioeconomic status.